ENTITY RELATIONSHIP DIAGRAM · BIGQUERY-PUBLIC-DATA.NEW_YORK_CITIBIKE

How the Citi Bike tables join

Trips reference a station twice (start and end). The station IDs in the two tables come from different systems, so a direct ID join returns **zero rows**. The fix is a small crosswalk table built once by name and location, then every trip joins through it.



Join recipe

```
FROM citibike_trips t
JOIN station_xref xs ON t.start_station_id = xs.trip_station_id
JOIN station_xref xe ON t.end_station_id   = xe.trip_station_id
LEFT JOIN citibike_stations ss ON xs.station_id = ss.station_id
LEFT JOIN citibike_stations es ON xe.station_id = es.station_id
```

Build the crosswalk from the **union** of start and end IDs (926 total). Match on `LOWER(TRIM(name))` first, then nearest station with `ST_DISTANCE` under 100 m. Keep LEFT JOINs: retired stations have no match.

Don't: `ON CAST(t.start_station_id AS STRING) = s.station_id` matches 0 of 881 stations.

Crow's foot = many · **bar** = exactly one · **circle + bar** = zero or one **Purple solid** = key join **Grey dashed** = attributes used to build the crosswalk (name, lat/long) **Faded rows** = not needed for joins

Relationships are logical; BigQuery enforces none of them · match rates from BigQuery on 2026-11-03